Agentic Workflows: Design and Implementation (online)

The next course dates are not yet confirmed. Add your name to the **waiting list** for a priority update on this course.

Sign up for **course news** to receive updates about our Technology and AI programme.

Overview

While ChatGPT was disruptive, 'autonomous AI agents' promise to be potentially even more revolutionary.

Essentially, autonomous AI agents allow you to instruct AI to perform tasks at a higher level of abstraction. For example, instead of providing detailed instructions, you can simply ask the AI to 'book a holiday to Greece' - the AI would then create and prioritise subtasks, and semi-autonomously execute them to achieve your goal.

This course covers the development of systems concerned with autonomous AI agents (agentic workflows).

Autonomous AI agents are large language models (LLM) based systems that are distinguished by ‘agency’. These systems can act semi-autonomously to interact with other systems, make decisions and perform a complex goal. In contrast, traditional systems are concerned with a transactional interaction i.e. a one pass engagement.

Components of agentic workflows include (as proposed by technology entrepeneur Andrew Ng):

- Reflection: the ability to examine and improve its own work
- Tool use: actuate tasks by invoking tools
- Planning and reasoning: develop and execute multi-step plans to achieve the goal (problem solving abilities)
- Multi-agent collaboration: the ability for multiple AI agents to work together to communicate and coordinate to solve a larger task

View all data science, computing and maths courses

View all technology and ai courses

Sign up for course news

Receive email updates about our Technology and AI CPD Courses

Programme details

The course is primarily concerned with redesigning enterprise workflows using autonomous AI agents.

In this course, we cover:

- Design and development of autonomous AI agents
- Agentic RAG (retrieval augmented generation)
- Designing agentic workflows
- Simulations at scale
- Specific tools and techniques for development of agentic workflows such as OpenAI, llamaindex, AWS and Azure
- Processes, standards and safety

Note that this is a complex and dynamic topic. Both the themes and speakers are subject to change as we approach the start of the course.

The course is positioned at both a strategic and a technical level. Code will be both used and demonstrated in the course, but participants will not be expected to write their own code as part of the course.

Dates, Times and Delivery

This course will run over six live online sessions on Mondays, Wednesdays and Fridays, from 14:00 - 18:00 (UK time).

In some cases, the sessions may extend to 18:30, and will be delivered online via Microsoft Teams.

A world clock, and time zone converter can be found here: https://bit.ly/3bSPu6D

No attendance at Oxford is required and you do not need to purchase any software.

Accessing Your Online Course

Details about accessing the private MS Teams course site will be emailed to you during the week prior to the course commencing.

Please get in touch if you have not received this information within three working days of the course start date.

Certification

In order to be eligible for a certificate, you will need to attend the whole course. Participants who meet this criterion will be emailed after the end of the course with a link, and instructions on how to access their University of Oxford digital certificate.

The certificate will show your name, the course title and the dates of the course you attended. You will be able to download your certificate, as well as share it on social media if you choose to do so.

Fees

Description	Costs
Course fee	£1440.00

Payment

Fees include electronic copies of course materials.

All courses are VAT exempt.

Register immediately online

Click the 'Book now' button on this webpage. Payment by credit or debit card is required.

Request an invoice

If you require an invoice for your company or organisation, please email us to request an online enrolment form. Payment is then accepted online, by credit/debit card, or by bank transfer.

Tutors

Ajit Jaokar - Course Director

Ajit is a dedicated leader and teacher in Artificial Intelligence (AI), with a strong background in AI for Cyber-Physical Systems, research, entrepreneurship, and academia.

Currently, he serves as the Course Director for several AI programs at the University of Oxford and is a Visiting Fellow in Engineering Sciences at the University of Oxford. His work is rooted in the interdisciplinary aspects of AI, such as AI integration with Digital Twins and Cybersecurity.

His courses have also been delivered at prestigious institutions, including the London School of Economics (LSE), Universidad Politécnica de Madrid (UPM), and as part of The Future Society at the Harvard Kennedy School of Government.

As an Advisory AI Engineer, Ajit specialises in developing innovative, early-stage AI prototypes for complex applications. His work focuses on leveraging interdisciplinary approaches to solve real-world challenges using AI technologies.

Ajit has shared his expertise on technology and AI with several high-profile platforms, including the World Economic Forum, Capitol Hill/White House, and the European Parliament.

Ajit is currently writing a book aimed at teaching AI through mathematical foundations at the high school level.

Ajit resides in London, UK, and holds British citizenship. He is actively engaged in advancing AI education and innovation both locally and globally. He is neurodiverse - being on the high functioning autism spectrum.

Ajit's work in teaching, consulting, and entrepreneurship is grounded in methodologies and frameworks he developed through his AI teaching experience. These methodologies help to rapidly develop complex, interdisciplinary AI solutions in a relatively short time. These include:
1. The Jigsaw Methodology for low-code data science to non-developers.
2. The AI Product Manager framework and AI product market fit framework
3. Software engineering with the LLM stack
4. Agentic RAG for cyber-physical systems.
5. AI for Engineering sciences:
6. The ability of AI to reason using large language models

He also consults at senior advisory levels to companies.

His newsletter on AI in Linkedin has a wide following
https://www.linkedin.com/newsletters/artificial-intelligence-6793973274368856064/

Anjali Jain - Tutor

Co-founder, Erdos | Author | Senior Tutor in AI & ML, University of Oxford | AI Ambassador, Oxford AI & ML Competency Center | Data Architect, Metro Bank

Anjali Jain is a London-based data architect, author, and AI expert with over two decades of experience in software development, architecture, data strategy, and applied machine learning.

At the University of Oxford, she serves as Senior Tutor in AI and Machine Learning and as AI Ambassador at the AI & ML Competency Center, where she leads strategic initiatives in AI education and research.

She is the co-founder of Erdos Research, a collaborative research and innovation lab focused on building and implementing AI systems, advancing prompting methods, and developing tools for AI-assisted software engineering.

Anjali also serves as Data Architect at Metro Bank, where she supports the integration of AI into financial systems with a focus on data governance, data architecture and compliance. She is the co-author of “10X AI Developer Guide with BRIDGE AI Framework” and “AI-Assisted Programming for Web and Machine Learning”, offering practical methodologies for building intelligent, human-centric technologies.

Abhinav Kimothi - Tutor

Abhinav Kimothi is a seasoned data science and AI leader with over 15 years of experience in data driven consulting, application development, and leveraging AI and ML to solve complex business problems. Presently, he is a co-founder and the Vice President of Artificial Intelligence at Yarnit where he leads a team dedicated to developing an innovative content marketing platform powered by generative AI.

Abhinav's career has spanned diverse projects in analytics, predictive modeling, machine learning and enterprise product development. Abhinav studied engineering at BITS-Pilani and got his management education at Indian School of Business – Hyderabad.

Passionate about driving AI advancements, he aims to make a meaningful impact by transforming data into actionable insights and pushing the boundaries of technology.

Nicole Königstein - Tutor

Nicole is the Co-Founder, CEO, and Co-Chief AI Officer at Quantmate, a deep-tech fintech company developing AI agents for portfolio management and strategy development via natural language. She is a globally recognized thought leader in large language models and agentic architectures, with a particular focus on their transformative applications in quantitative finance.

As a guest lecturer, Nicole shares her expertise in Python, machine learning, and deep learning at universities. She is also a frequent speaker at AI and quantitative finance events.

Nicole has authored Math for Machine Learning and Transformers in Action with Manning Publications. Her forthcoming book, Transformers: The Definitive Guide – Applications Beyond NLP, will be published by O’Reilly Media.

Christoffer Noring - Tutor

Senior Cloud Advocate, Microsoft

Chris is Senior Cloud Advocate at Microsoft with more than 15 years's experience in the IT industry. He's a published author on several books about web development as well as the Go language. He's also a recognized speaker as well as keynote speaker and holds a Google developer expert title.

Arthur Orts - Tutor

Arthur Orts is Head of Generative and Agentic AI at Pictet Asset Management in Geneva, where he leads the development and adoption of AI capabilities across the firm.
Prior to joining Pictet, Arthur spent 10 years at Bloomberg in London, working in Financial Solutions with a focus on buy-side clients and portfolio management/risk analytics tools.

Arthur combines deep expertise in financial technology with hands-on experience building AI-powered workflows and agentic systems. He is particularly focused on the practical implementation of Agentic Artificial intelligence into professional environments, including coding assistants, autonomous agents and enterprise AI adoption strategies.

Claudia Saleh - Tutor

Claudia Saleh is an AI Product Leader at Disney with over 20 years of IT experience. She traded the laid-back beaches and sunny Rio de Janeiro, Brazil, where she worked for media companies like Globo.com and ADVPress, for the dynamic international scene of Washington, DC. There, she contributed her expertise to international organisations such as the World Bank, the Inter-American Development Bank, and the United Nations.

She works in the media and entertainment industry, driving innovation at the intersection of technology and creativity. As a graduate student in Artificial Intelligence, she combines academic insights with hands-on expertise, focusing on AI strategies and their transformative potential for knowledge and creative professionals.

An experienced speaker and mentor, Claudia has guided enterprises in adopting technology effectively and strives to empower professionals to see AI as a collaborator. Her diverse background includes a decade as a travel journalist and graphic designer, which adds a unique perspective to her work, enabling her to simplify complex ideas and inspire diverse audiences.

When she’s not exploring the latest AI trends, Claudia can be found writing, travelling, or immersing herself in the magic of Disney. She brings a unique voice to conversations about AI, blending technical insights with a deep appreciation for the creative spark that drives innovation.

Richard Taylor - Tutor

Richard Taylor is an AI/ML Lead with over a decade of experience in machine learning and AI within the financial sector. He has a background in Astrophysics, specialising in Computational Cosmology at the University of Manchester, where he conducted statistical analysis of large-scale universe simulations.

Before joining Ninety One, Richard was a Senior Data Scientist at Fable Data, where he was the first technical hire. He built the company’s initial data pipelines, machine learning models, and MVP, securing major investment clients and driving business growth. He also led and expanded the Product and Compliance functions within the Data Science division, fine-tuning Large Language Models for near real-time data classification and automating the capture and processing of additional datasets using AI.

Earlier in his career, he worked at Barclays Bank in Market Surveillance, developing machine learning models to monitor trading activity, following time spent working for KPMG. Now, at Ninety One, he is responsible for looking at applications of AI across the business with a focus on investments.

Application

If you would like to discuss your application or any part of the application process before applying, please click the 'Ask a Question' button at the top of this page.

IT requirements

This course is delivered online using Microsoft Teams. You will be required to follow and implement the instructions we send you to fully access Microsoft Teams on the University of Oxford's secure IT network.

This course is delivered online; to participate you will need regular access to the Internet and a computer meeting our recommended *Minimum computer specification.*

It is advised to use headphones with working speakers and microphone.

Terms & conditions
for applicants and students

Information on
financial support

Contact us

Oxford Lifelong Learning at the Department for Continuing Education
University of Oxford
Rewley House, 1 Wellington Square
Oxford
OX1 2JA
+44 (0)1865 270360

Find us

Contact us

Stay informed

Information

About us

Online and virtual courses

Day and weekend classes

Weekly learning courses

Professional development

Oxford qualifications

Summer schools

Academic staff profiles

Support lifelong learning

Jobs and vacancies

Terms and conditions

Student and staff

Student support

Using the library

Using Moodle

Canvas login

Nexus login

University of Oxford | Privacy policy | How we use cookies | Website feedback | Accessibility Statement

Stay in touch